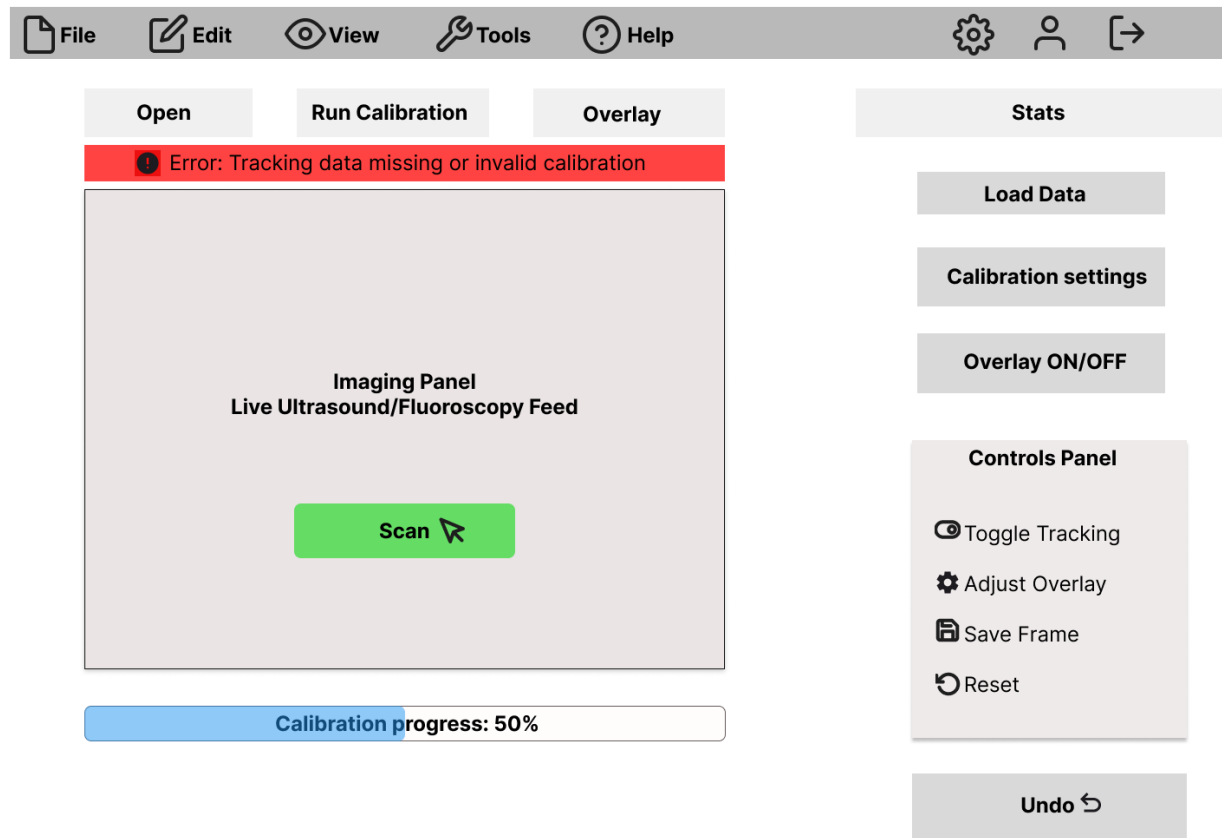


UX Artifacts – Wireframes, Personas, User Journeys

Wireframe 1 – Main imaging and Tracking Screen



Wireframe 2 – Calibration Workflow

File

Edit

View

Tools

Help

Calibration Panel

Load ROM File

X coordinate:

Y coordinate:

Z coordinate:

Error value:

✓ Calibration Successful

Calibration progress: 100%

Controls Panel

X = 123.6, Y = 367.8, Z = 506.3
Error value = 1.01

Save Calibration

Reset

Retry

Wireframe 3 – 3D Reconstruction Panel

File

Edit

View

Tools

Help

3D Reconstruction Visualization

❗ Frame count below minimum threshold for 3D Reconstruction

Drag to Rotate 3D Volume

Reconstruction in progress: 40%

Frames loaded: 14/128

Estimated time
till completion: 10m 31s

Reconstruct 3D Volume

Reconstruct Slice

Opacity Slider

Save Slice

Undo last reconstruction

Export

UX Artifacts: Personas

1st Persona:

Persona Name	Surgical Resident
Tech Level	Skills: Medium – Clinically trained with standard tools, has no advanced coding knowledge.
Role	A user who employs the system for training and procedural purposes along with imaging overlays.
Goal	• Be able to calibrate the devices in a quick and efficient manner. • Utilize ultrasound + 3D reconstruction streaming for real-time guidance of the simulation.
Pain Points	• Calibration most of the time fails and no reasons are given. • Reconstruction is too lengthy → breaks the workflow. • It is difficult to distinguish whether the calibration is “done” or “failed.”
Accessibility	• Unambiguous visual indications (green = success, red = error). • Big, labeled buttons with the least possible number of clicks. • A simple retry path for a scan that has failed.

2nd Persona:

Persona Name	Biomedical engineer
Tech Level	Skills: High - knows Python, C++, system integration.
Role	Safeguards backend, checks algorithms of reconstruction, incorporates additional modules.
Goal	• Make sure that there is modularity and testability in the calibration and reconstruction modules. • Fix bugs in the debug system that can have repeatable logs.
Pain Points	• Code is strongly coupled, and it is difficult to update. • The systems that existed previously did not have proper logging. • It is difficult to test errors that are not standardized tests.
Accessibility	• The debug logs and error codes accessible to the developer. • API documentation. • Ability to inject test data.

3rd Persona:

Persona Name	Clinician
Tech Level	Skills: Low - minimum UI interaction.
Role	Oversees procedures, must have validations of reconstructions, clinical decisions.
Goal	• Calibration of trust and 3D reconstructions in order to make accurate choices. • Reduce operations time delays.
Pain Points	• Loses confidence in case of unclear UI. • Is unable to afford the downtime of the system in the middle of a procedure. • Wants assurance that reconstruction is a faithful one.
Accessibility	• Simplified, high-contrast UI. • Automatic notices (e.g., "Calibration Complete," "Reconstruction Saved" etc.). • Not a single manual troubleshooting is necessary.

UX Artifacts: User Journeys

Journey 1 – Calibrating Workflow (Resident + Engineer)

User initiates software → picks Calibration → Loads ROM file of stylet/camera → Calibration of the system is attempted.

- Success: Green banner "Calibration complete." Failure: Red banner + directed retrieval choice.
- Debug View can be opened by engineer to view logs.
- Resident receives the updated values of coordinates (x, y, z, error values) in live feed.
- **Edge case:** ROM file does not exist - brings user to manual fields of entry.

Journey 2 – Imaging and 3D Reconstruction (Resident + Clinician)

Resident begins ultrasound feeding → Sequential frames are automatically captured in a system → The user clicks on Reconstruct 3D Volume → Progress bar and preview window display reconstructing progress → Clinician regulates the visualization using the opacity/contrast → System compares reconstructed volume and baseline.

- **Edge case:** Not enough frames captured → banner read as "Frame set incomplete. Please capture more frames."

Journey 3 – Surgeon – Clinical Workflow Logging

Surgeon begins operation → events are automatically recorded by the system → important milestones noted (Tool identified → Calibration completed → Reconstruction stored) → On conclusion of session- surgeon clicks Export Log and Results → System produces ordered report containing timestamps and volume file that has been saved in 3D

Edge case: The system crashes in the middle of the session, but autosave quickly restores the earlier successful state of calibration/reconstruction.